

$$1234 \times 5678$$

$$X = 1234 \Rightarrow a \times 10^n + b$$

$$\Rightarrow 12 \times 10^2 + 34$$

$$Y = 5678 \Rightarrow c \times 10^m + d$$

$$\Rightarrow 56 \times 10^2 + 78$$

$$a = 12 \times 10^2, b = 34, c = 56 \times 10^2, d = 78, n = 2$$

$$a * c \Rightarrow 12 \times 10^2 * 56 \times 10^2$$

$$\Rightarrow 672 \times 10^4$$

$$b * d \Rightarrow 34 * 78$$

$$\Rightarrow 2652$$

$$a * d \Rightarrow 12 \times 10^2 * 78$$

$$\Rightarrow 936 \times 10^2$$

$$b * c \Rightarrow 34 * 56 \times 10^2$$

$$\Rightarrow 1904 \times 10^2$$

$$\Rightarrow (a * c) + (b * d) + (b * c) + (a * d)$$

$$\Rightarrow 672 \times 10^4 + 2652 + 1904 \times 10^2 + 936 \times 10^2$$

$$\Rightarrow 6720000 + 2652 + 190400 + 93600$$

$$\Rightarrow 7006652$$

$$x = 41, \quad y = 42$$

$$x = a \times 10^n + b$$

$$x = 4 \times 10^1 + 1$$

$$y = c \times 10^n + d$$

$$y = 4 \times 10^1 + 2$$

$$a = 4 \times 10^1, b = 1, c = 4 \times 10^1, d = 2$$

$$(a * c) \Rightarrow 4 \times 10^1 * 4 \times 10^1$$

$$\Rightarrow 16 \times 10^2$$

$$(b * d) \Rightarrow 1 \times 2 \Rightarrow 2$$

$$(a * d) \Rightarrow 4 \times 10^1 * 2$$

$$\Rightarrow 8 \times 10^1$$

$$(b * c) \Rightarrow 1 * 4 \times 10^1$$

$$\Rightarrow 4 \times 10^1$$

$$\Rightarrow (a * c) + (b * d) + (a * d) + (b * c)$$

$$\Rightarrow (16 \times 10^2) + 2 + 8 \times 10^1 + 4 \times 10^1$$

$$\Rightarrow 1600 + 2 + 80 + 40$$

$$\Rightarrow 1722$$